mAP for beam search

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The outline for computing the precision-recall curve and mAP is as follows:

1. Make a list of all predictions. Confidence of a prediction is the log-probability of the beam under the model:
   `pred = [(episode_i, conf_i, traj_i)]; len(pred) = N_pred`
   Make a list of true trajectories:
   `true = [(episode_i, 1.0, traj_i)]; len(true) = N_true`
2. Sort predictions by descending confidence
3. For each predicted trajectory:
   (a) calculate the L1 error with the true trajectory from the same episode
   (b) if `error > error_thr` or another prediction was already found for this true trajectory, it is a false positive
   (c) otherwise, the prediction is a true positive
4. Calculate `precision_i` and `recall_i` for the first `i` predictions (which is equivalent to ignoring all predictions with `conf_j < conf_i`). Precision is `N_TP/N_pred`, recall is `N_TP/N_true`.
5. Plot the (recall, precision) points and compute Area Under Curve (AUC)
6. Repeat the AUC calculation for several L1 error thresholds, the average AUC is the mean Average Precision

1 References

1. Jonathan Hui
2. code
3. learnopencv - additional theory
4. pyimagesearch - additional theory